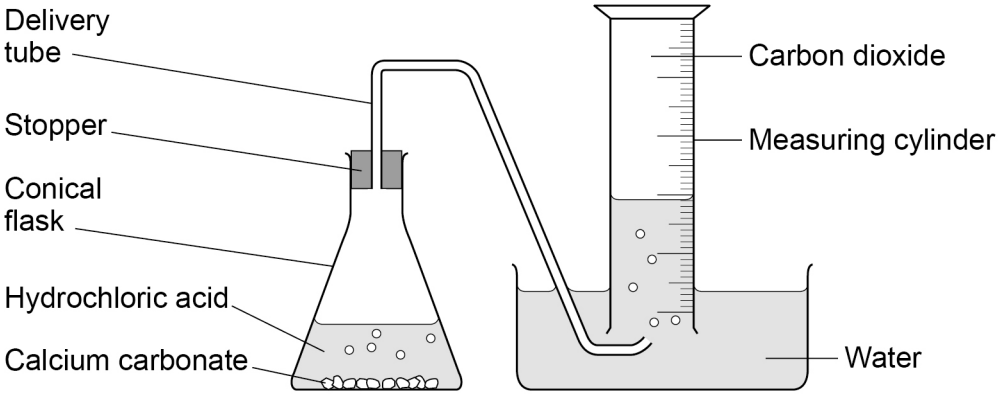
Do not write outside the box

**0 2** Some factors affect the rates of chemical reactions.

**0 2 . 1** A student investigated the effect of changing the particle size of calcium carbonate on the rate of reaction with hydrochloric acid.

**Figure 3** shows the apparatus.

**Figure 3**




Describe a method the student could use to produce valid results.

**[6 marks]**

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**0 2 . 2** The student investigated the effect of increasing the temperature on the rate of a reaction.

Explain the effect of increasing the temperature on the rate of a reaction.

Refer to particles and collisions in your answer.

**[3 marks]**

Do not write outside the box

Catalysts affect the rate of reactions.

**0 2 . 3** What is meant by a 'catalyst'?

**[2 marks]**

_____

_____

_____

_____

**0 2 . 4** What are catalysts in biological systems called?

**[1 mark]**

_____

12

**Turn over for the next question**